

Zhenkang Xu

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EDUCATION

Western University

London, ON

B.Sc. (Honours Specialization in Computer Science) | Software Engineering

2023 – Present

- **GPA:** 3.60/4.00
- **Relevant courses:** Algorithm Design, Data Structures, Operating Systems, Computer Organization, Software Engineering, Web Programming, Artificial Intelligence, Database, Project Management

SKILLS

- **Programming Languages:** C++, Python (Django, Flask, NumPy, Pandas, PyTorch), Java (Swing, Spring Boot), SQL (PostgreSQL), HTML, CSS, JavaScript (ReactJS, NextJS), TypeScript, Bash, R
- **Services & Tools:** Git, Docker, Kubernetes, AWS, PowerBI

PROJECT EXPERIENCE

Pre-IPO Markpet | Elixir, Phoenix, React, TypeScript, PostgreSQL, WebSocket

Full Stack Developer

- Built a **full-stack** order matching engine with price-time priority algorithm, supporting partial fills within ACID transactions
- Implemented real-time order book and trade feed updates via **Phoenix Channels (WebSocket)**, enabling multi-user live trading
- Designed **RESTful API (8 endpoints)** with structured **Ecto schemas** for users, orders, trades, and holdings with referential integrity
- Developed **React** frontend with **TanStack Query** for server state management and custom hooks for **WebSocket** channel subscriptions

Virtual Pet Desktop Game | Java, Swing, Gson, JUnit 5

Team Lead, System Architecture Designer

- Engineered a pet state machine with 5 states (Normal/Hungry/Angry/Sleeping/Dead) and type-specific decay rates, driving real-time stat updates via **Swing Timer**
- Built the core gameplay screen with live progress bars, **WebSocket-style** status sync, and warning indicators for critically low stats
- Implemented cooldown-based interaction system for play and vet actions, plus a threaded sleep mechanic with automatic state transitions
- Designed pet selection and main menu UI with dynamic image rendering, pet naming dialog, and navigation across 6+ screens
- Developed feed/gift consumption logic with max-cap validation, score tracking, and inventory integration in a 5-person team

Western Scheduler App | C++, Qt 6, Google Test/Mock, JSON

Team Lead, backend developer

- Architected a desktop calendar app using **MVC** pattern with Observer (Signal/Slot), Factory, and Prototype **design patterns**
- Implemented recurring event engine that dynamically expands weekly course patterns via ISO day-of-week calculation and clone-based instantiation
- Built custom QCalendarWidget with owner-drawn event indicators, supporting color-coded dots and overflow display for dense schedules
- Developed Controller layer managing event lifecycle, **JSON** persistence via LocalFileStorage, and a 5-second background reminder timer
- Validated edge cases (null pointers, boundary dates, simulated backend failures) using **Google Test**.

PROFESSIONAL EXPERIENCE

ItemLookBook

Vancouver, BC

Associate Software Engineer

May 2022 – Aug 2023

- Collaborated with team to manage 1,000+ product listings and inventory across website and mobile app using **JavaScript** and **SQL**, maintaining accurate stock levels and product data.
- Co-developed a batch update script, reducing manual product pricing and inventory update time by **60%** across 30+ listings.

TopDance Studio

Vancouver, BC

Founder & Software Developer

May 2024 – Aug 2025

- Built and deployed a full-stack student management web app using **React**, **Node.js**, and **PostgreSQL**, supporting 100+ students with features including attendance tracking, credit management, and course scheduling.
- Grew Rednote presence to **2,000+ followers**, achieved **70,000+ views** on a single video, driving an **80% increase** in student enrollment.